

Aayush Puri
AI & ML Engineering | B.Tech Computer Science
Ghaziabad, UP
Dr. A.P.J. Abdul Kalam Technical University

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Professional Summary

AI/ML Engineer with 1 year of hands-on experience in developing and deploying machine learning and deep learning solutions. Skilled in computer vision, basic NLP applications, model optimization, and edge deployment. Proficient in Python, PyTorch, TensorFlow, and NVIDIA Jetson platforms. Currently working as an AI Developer at Itdose Pvt. Ltd.

Education

•Bachelor of Technology in Computer Science
Dr. A.P.J. Abdul Kalam Technical University
Graduation: 2025
Specialization: AI & ML

Professional Experience

•AI Developer
– Developing and implementing AI solutions
Sep 2025 - Present
Itdose Pvt. Ltd. - India
Dec 2024 - Sep 2025

•AI & Machine Learning Engineer
– Developed deep learning pipelines for real-time drone video analytics and navigation support.
– Optimized YOLO-based object detection models with TensorRT on NVIDIA Jetson AGX Orin, achieving ~30% faster inference.
– Implemented stereo depth estimation for obstacle detection and 3D environment understanding.
– Integrated LLM-based interfaces for natural language drone command processing.
– Deployed computer vision pipelines on ROS2-enabled Jetson edge devices for real-time inference.
Enhanced detection performance and reduced latency through model optimization and edge deployment techniques.
Enord - Delhi, India

Key Projects

•Custom Whisper for Hinglish Recognition
– Fine-tuned Whisper model for Hinglish (Hindi-English) speech-to-text recognition
– Achieved 85% accuracy for mixed-language audio inputs
– Deployed for drone command systems enabling natural language control

•YOLOv11m Multi-Class Detection Pipeline
– Trained on 24K+ images for people, bird, and FOD (Foreign Object Debris) detection
– Achieved mAP@0.5 of 0.92 for real-time crowd monitoring applications
– Optimized for deployment on edge devices with minimal latency

LLM-Based Drone Command Interface
– Fine-tuned a LLaMA model for natural language understanding of drone control commands
– Developed a system to translate text instructions into structured flight control parameters
– Integrated the LLM module with the drone software stack for command interpretation

Technical Skills

Programming Python
ML/DL Frameworks : PyTorch, TensorFlow
Computer Vision: OpenCV, YOLO,
NLP & AI : Whisper, LLMs, VLMs, Generative AI
Data Processing NumPy, Pandas, Matplotlib
Deployment ONNX, TensorRT, DeepStream

Certifications

•AI & ML Engineering: Enord 2024